# Relatório Técnico - Laboratório MPLS, VPLS e OSPF no MikroTik

Data: 07/08/2026

Mauricio
Junior

# Requisitos:

- **Infraestrutura e Roteamento (Backbone):**
    - Executar a rede utilizando roteadores MikroTik.
    - Rede em MPLS.
    - Aplicar de forma idêntica (simétrica) os custos definidos para as conexões em ambos os lados de cada enlace.

- **Serviço L2VPN (Túnel L2):**
    - Estabelecer um túnel de Camada 2 (L2) entre o R8 e o R1.
    - Conectar um equipamento em cada extremidade desse túnel (nos switches associados ao R1 e R8) e garantir a comunicação direta em L2 entre eles.

- **Servidor e Aplicações:**
    - Adicionar um servidor Ubuntu à topologia.
    - Deixar o servidor plenamente acessível pela rede e com o serviço web Nginx instalado e rodando.

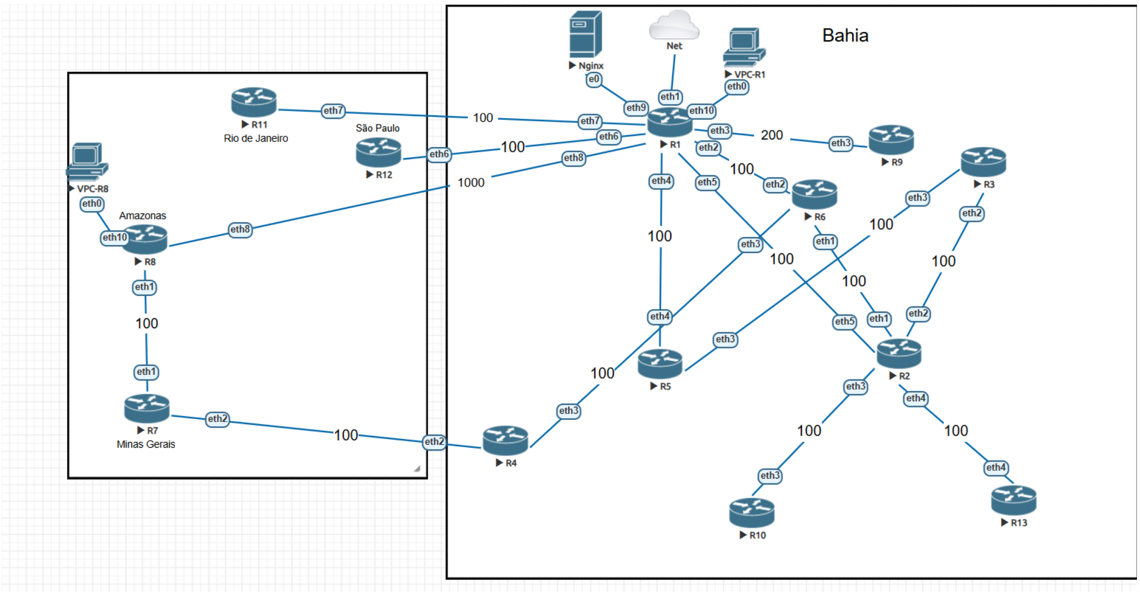
- **Evidências e Documentação:**
    - Capturar prints das adjacências estabelecidas em todos os equipamentos da rede.
    - Executar e registrar testes de `ping` e `traceroute` entre o R3 e o R7 em ambas as direções (R3 -> R7 e R7 -> R3).

- **Gerenciamento:**
    - Efetuar o backup completo das configurações de todos os roteadores para permitir restauração rápida caso necessário.

# 1.Topologia:

- **Visão Geral:** A rede é composta por 13 Routers Mikrotik compartilhando rotas via OSPF com métricas de custo e autenticação definidos. O MPLS está sendo usado para o encaminhamento de tráfego, também foi estabelecido uma VPLS entre R1 e R8 para que os equipamentos da Lan se comunicassem via Layer 2, atribuindo regras de firewall para que não houvesse comunicação da Lan com os Routers pelos endereços de Loopback ou pelos endereços estabelecidos nas interfaces físicas dos routers. Simulando uma aplicação, foi adicionado um servidor Linux com Nginx que está acessível por toda a rede, incluindo a Lan do R1 e do R8. Por fim, todos os dispositivos conseguem se comunicar com a internet.

- **Diagrama da Rede:**




## 2.Nomeclaturas:

| | |
|---|---|
| • R1-13 | • Roteadores da infraestrutura numerados de 1 a 13 |
| • VPC-R1-R8 | • Máquinas do túnel VPLS R1 e R8 |
| • Net | • Conexão com a Internet |
| • Nginx | • Aplicação Web |
| • 100;200;1000 | • Custos OSPF |
| • Estados | • Indica a região onde estão localizados os equipamentos |

# 3.Ligação das Interfaces:

Para facilitar a gerência e a configuração dos equipamentos, a ligação física dos Routers seguiu o seguinte padrão: Ambas as extremidades utilizam a mesma numeração, criando uma simetria. R1 ether2 : R6 ether2

| | | | | |
|---|---|---|---|---|
| **R1** | **R8** | | **R2** | **R13** |
| **Ether8** | **Ether8** | | **Ether4** | **Ether4** |
| **R1** | **R6** | | **R2** | **R10** |
| **Ether2** | **Ether2** | | **Ether3** | **Ether3** |
| **R1** | **R11** | | **R2** | **R3** |
| **Ether7** | **Ether7** | | **Ether2** | **Ether2** |
| **R1** | **R12** | | **R2** | **R6** |
| **Ether6** | **Ether6** | | **Ether1** | **Ether1** |
| **R1** | **R9** | | **R6** | **R4** |
| **Ether3** | **Ether3** | | **Ether3** | **Ether3** |
| **R1** | **R5** | | **R5** | **R3** |
| **Ether4** | **Ether4** | | **Ether3** | **Ether3** |
| **R1** | **R2** | | **R7** | **R4** |
| **Ether5** | **Ether5** | | **Ether2** | **Ether2** |
| **R8** | **R7** | | **R1** | **Nginx** |
| **Ether1** | **Ether1** | | **Ether9** | **Ether0** |
| **R1** | **VPC-R1** | | **R8** | **VPC-R8** |
| **Ether10** | **Ether0** | | **Ether10** | **Ether0** |
| **R1** | **Internet** | | | |
| **Ether1** | | | | |

## 4. Infraestrutura de Roteamento e Transporte:

Para roteamento dinâmico, o OSPF foi implementado no Backbone, tendo como Router ID o endereço de Loopback. Dessa forma, é garantido uma melhor gerência da rede e uma conectividade estável, pois, a interface lógica de Loopback não ficará Down.

- **OSPF Adjacências:**

```
[admin@R1] > routing/ospf/neighbor/print

Flags: V - virtual; D - dynamic

 0  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.65

      router-id=10.100.0.11 state="Full" state-changes=5 adjacency=3m31s

      timeout=39s


 1  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.30

      router-id=10.100.0.2 state="Full" state-changes=4 adjacency=2m57s

      timeout=34s


 2  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.61

      router-id=10.100.0.12 state="Full" state-changes=5 adjacency=3m31s

      timeout=39s


 3  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.34

      router-id=10.100.0.5 state="Full" state-changes=5 adjacency=3m33s

      timeout=37s


 4  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.2

      router-id=10.100.0.6 state="Full" state-changes=5 adjacency=3m33s

      timeout=37s


 5  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.6

      router-id=10.100.0.9 state="Full" state-changes=5 adjacency=3m36s

      timeout=34s


 6  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.53

      router-id=10.100.0.8 state="Full" state-changes=10 adjacency=2m11s

      timeout=35s
```

```
[admin@R2] > routing/ospf/neighbor/print
Flags: V - virtual; D - dynamic
 0  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.17
      router-id=10.100.0.6 state="Full" state-changes=5 adjacency=4h47m37s
      timeout=34s

 1  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.13
      router-id=10.100.0.3 state="Full" state-changes=6 adjacency=4h17m44s
      timeout=33s

 2  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.22
      router-id=10.100.0.10 state="Full" state-changes=6 adjacency=4h17m44s
      timeout=31s

 3  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.26
      router-id=10.100.0.13 state="Full" state-changes=6 adjacency=4h17m43s
      timeout=38s

 4  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.29
      router-id=10.100.0.1 state="Full" state-changes=4 adjacency=6m51s
      timeout=39s
```

```
[admin@R3] > routing/ospf/neighbor/print
Flags: V - virtual; D - dynamic
 0  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.14
      router-id=10.100.0.2 state="Full" state-changes=5 adjacency=4h18m29s
      timeout=34s

 1  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.9
      router-id=10.100.0.5 state="Full" state-changes=5 adjacency=4h18m25s
      timeout=38s
```

```
[admin@R4] > routing/ospf/neighbor/print
Flags: V - virtual; D - dynamic
 0  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.46
      router-id=10.100.0.7 state="Full" state-changes=6 adjacency=4h19m36s
      timeout=34s

 1  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.42
      router-id=10.100.0.6 state="Full" state-changes=5 adjacency=4h19m37s
      timeout=34s
```

```
[admin@R5] > routing/ospf/neighbor/print
Flags: V - virtual; D - dynamic
 0  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.10
      router-id=10.100.0.3 state="Full" state-changes=6 adjacency=4h20m6s
      timeout=37s

 1  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.33
      router-id=10.100.0.1 state="Full" state-changes=11 adjacency=9m53s
      timeout=33s
```

```
[admin@R6] > routing/ospf/neighbor/print
Flags: V - virtual; D - dynamic
 0  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.18
      router-id=10.100.0.2 state="Full" state-changes=6 adjacency=4h50m36s
      timeout=31s

 1  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.1
      router-id=10.100.0.1 state="Full" state-changes=11 adjacency=10m24s
      timeout=31s

 2  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.41
      router-id=10.100.0.4 state="Full" state-changes=6 adjacency=4h20m45s
      timeout=35s
```

```
[admin@R7] > routing/ospf/neighbor/print
Flags: V - virtual; D - dynamic
 0  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.50
      router-id=10.100.0.8 state="Full" state-changes=11 adjacency=9m50s
      timeout=31s

 1  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.45
      router-id=10.100.0.4 state="Full" state-changes=5 adjacency=4h21m39s
      timeout=40s
```

```
[admin@R8] > routing/ospf/neighbor/print
Flags: V - virtual; D - dynamic
 0  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.49
      router-id=10.100.0.7 state="Full" state-changes=5 adjacency=10m59s
      timeout=31s

 1  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.54
      router-id=10.100.0.1 state="Full" state-changes=5 adjacency=11m4s
      timeout=36s
```

```
[admin@R9] > routing/ospf/neighbor/print
Flags: V - virtual; D - dynamic
 0  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.5
      router-id=10.100.0.1 state="Full" state-changes=10 adjacency=13m48s
      timeout=31s
```

```
[admin@R10] > routing/ospf/neighbor/print
Flags: V - virtual; D - dynamic
 0  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.21
      router-id=10.100.0.2 state="Full" state-changes=5 adjacency=4h24m45s
      timeout=39s
```

```
[admin@R11] > routing/ospf/neighbor/print

Flags: V - virtual; D - dynamic

 0  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.66

      router-id=10.100.0.1 state="Full" state-changes=11 adjacency=15m9s

      timeout=35s
```

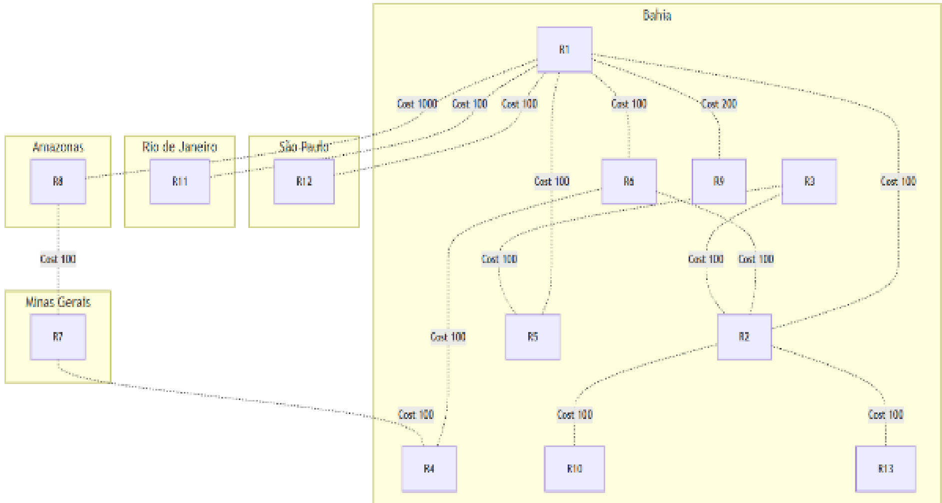
```
[admin@R12] > routing/ospf/neighbor/print

Flags: V - virtual; D - dynamic

 0  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.62

      router-id=10.100.0.1 state="Full" state-changes=11 adjacency=15m39s

      timeout=35s
```

```
[admin@R13] > routing/ospf/neighbor/print

Flags: V - virtual; D - dynamic

 0  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.25

      router-id=10.100.0.2 state="Full" state-changes=5 adjacency=4h26m23s

      timeout=30s
```

- ## Validação dos custos de interface

  O enlace R1-R8, com custo 1000, permanece como o caminho de menor preferência, dando a possibilidade de validar a seleção de caminhos e a redundância.

  **Os custos exigidos para o projeto estão listados na imagem abaixo:**



```
[admin@R1] > routing/ospf/interface-template/print
Flags: X - disabled, I - inactive
 0   area=ospf-area-1 interfaces=loopback,ether9-Nginx instance-id=0
     type=broadcast retransmit-interval=5s transmit-delay=1s
     hello-interval=10s dead-interval=40s priority=128 cost=1 passive
     auth=md5 auth-key="@Dm1122ti#"

 1   ;;; DEMAIS
     area=ospf-area-1
     interfaces=ether2-R6,ether4-R5,ether5-R2,ether6-R12,ether7-R11
     instance-id=0 type=ptp retransmit-interval=5s transmit-delay=1s
     hello-interval=10s dead-interval=40s priority=128 cost=100 auth=md5
     auth-key="@Dm1122ti#"

 2   ;;; R9
     area=ospf-area-1 interfaces=ether3-R9 instance-id=0 type=ptp
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=200 auth=md5 auth-key="@Dm1122ti#"

 3   ;;; R8
     area=ospf-area-1 interfaces=ether8-R8 instance-id=0 type=ptp
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=1000 auth=md5 auth-key="@Dm1122ti#"
```

```
[admin@R2] > routing/ospf/interface-template/print
Flags: X - disabled, I - inactive
 0   area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=1 passive auth=md5
     auth-key="@Dm1122ti#"

 1   ;;; DEMAIS
     area=ospf-area-1
     interfaces=ether1-R6,ether2-R3,ether3-R10,ether4-R13,ether5-R1
     instance-id=0 type=ptp retransmit-interval=5s transmit-delay=1s
     hello-interval=10s dead-interval=40s priority=128 cost=100 auth=md5
     auth-key="@Dm1122ti#"
```

```
[admin@R3] > routing/ospf/interface-template/print
Flags: X - disabled, I - inactive
 0   area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=1 passive auth=md5
     auth-key="@Dm1122ti#"

 1   area=ospf-area-1 interfaces=ether2-R2,ether3-R5 instance-id=0 type=ptp
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=100 auth=md5 auth-key="@Dm1122ti#"
```

```
[admin@R4] > routing/ospf/interface-template/print
Flags: X - disabled, I - inactive
 0   area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=1 passive auth=md5
     auth-key="@Dm1122ti#"

 1   ;;; DEMAIS
     area=ospf-area-1 interfaces=ether2-R7,ether3-R6 instance-id=0 type=ptp
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=100 auth=md5 auth-key="@Dm1122ti#"
```

```
[admin@R5] > routing/ospf/interface-template/print
Flags: X - disabled, I - inactive
 0   area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=1 passive auth=md5
     auth-key="@Dm1122ti#"

 1   ;;; DEMAIS
     area=ospf-area-1 interfaces=ether3-R3,ether4-R1 instance-id=0 type=ptp
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=100 auth=md5 auth-key="@Dm1122ti#"
```

```
[admin@R6] > routing/ospf/interface-template/print
Flags: X - disabled, I - inactive
 0   area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=1 passive auth=md5
     auth-key="@Dm1122ti#"

 1   ;;; DEMAIS
     area=ospf-area-1 interfaces=ether1-R2,ether2-R1,ether3-R4 instance-id=0
     type=ptp retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=100 auth=md5 auth-key="@Dm1122ti#"
```

```
[admin@R7] > routing/ospf/interface-template/print
Flags: X - disabled, I - inactive
 0   area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=1 passive auth=md5
     auth-key="@Dm1122ti#"

 1   ;;; DEMAIS
     area=ospf-area-1 interfaces=ether1-R8,ether2-R4 instance-id=0 type=ptp
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=100 auth=md5 auth-key="@Dm1122ti#"
```

```
[admin@R8] > routing/ospf/interface-template/print
Flags: X - disabled, I - inactive
 0   area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=1 passive auth=md5
     auth-key="@Dm1122ti#"

 1   ;;; R7
     area=ospf-area-1 interfaces=ether1-R7 instance-id=0 type=ptp
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=100 auth=md5 auth-key="@Dm1122ti#"

 2   ;;; R1
     area=ospf-area-1 interfaces=ether8-R1 instance-id=0 type=ptp
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=1000 auth=md5 auth-key="@Dm1122ti#"
```

```
[admin@R9] > routing/ospf/interface-template/print
Flags: X - disabled, I - inactive
 0   area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=1 passive auth=md5
     auth-key="@Dm1122ti#"

 1   ;;; R1
     area=ospf-area-1 interfaces=ether3-R1 instance-id=0 type=ptp
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=200 auth=md5 auth-key="@Dm1122ti#"
```

```
[admin@R10] > routing/ospf/interface-template/print
Flags: X - disabled, I - inactive
 0   area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=1 passive auth=md5
     auth-key="@Dm1122ti#"

 1   ;;; R2
     area=ospf-area-1 interfaces=ether3-R2 instance-id=0 type=ptp
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=100 auth=md5 auth-key="@Dm1122ti#"
```

```
[admin@R11] > routing/ospf/interface-template/print
Flags: X - disabled, I - inactive
 0   area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=1 passive auth=md5
     auth-key="@Dm1122ti#"

 1   ;;; R1
     area=ospf-area-1 interfaces=ether7-R1 instance-id=0 type=ptp
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=100 auth=md5 auth-key="@Dm1122ti#"
```

```
[admin@R12] > routing/ospf/interface-template/print
Flags: X - disabled, I - inactive
 0   area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=1 passive auth=md5
     auth-key="@Dm1122ti#"

 1   ;;; R1
     area=ospf-area-1 interfaces=ether6-R1 instance-id=0 type=ptp
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=100 auth=md5 auth-key="@Dm1122ti#"
```

```
[admin@R13] > routing/ospf/interface-template/print
Flags: X - disabled, I - inactive
 0   area=ospf-area-1 interfaces=loopback instance-id=0 type=broadcast
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=1 passive auth=md5
     auth-key="@Dm1122ti#"

 1   ;;; R2
     area=ospf-area-1 interfaces=ether4-R2 instance-id=0 type=ptp
     retransmit-interval=5s transmit-delay=1s hello-interval=10s
     dead-interval=40s priority=128 cost=100 auth=md5 auth-key="@Dm1122ti#"
```

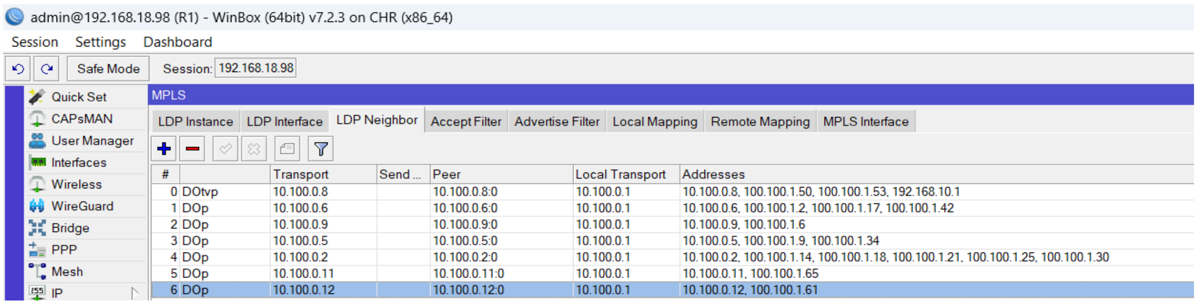
- **MPLS**

  O MPLS foi habilitado sobre a infraestrutura do OSPF e utiliza como Endereço de Transporte a Loopback, dessa forma, é garantido uma melhor gerência da infraestrutura e evita perdas de conectividade causados por problemas nos enlaces físicos.

Flags MPLS

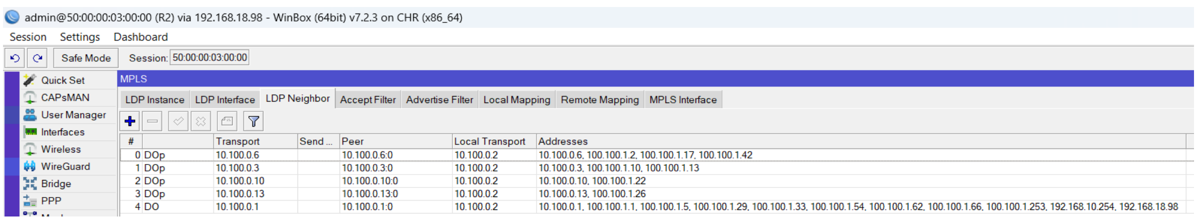
| | |
|---|---|
| D | Relação criada dinamicamente |
| O | Operante |
| T | Troca de mensagem Unicast com R8 |
| V | Conexão VPLS |
| P | Atuando como passivo |

R1



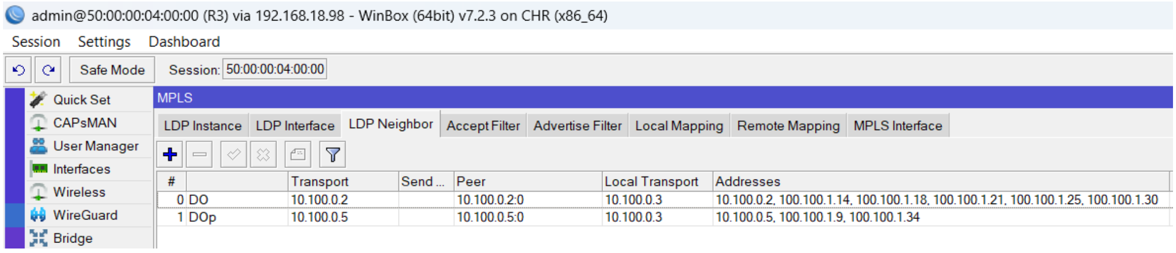

| # | | Transport | Send ... | Peer | Local Transport | Addresses |
|---|---|---|---|---|---|---|
| 0 | DOtvp | 10.100.0.8 | | 10.100.0.8:0 | 10.100.0.1 | 10.100.0.8, 100.100.1.50, 100.100.1.53, 192.168.10.1 |
| 1 | DOp | 10.100.0.6 | | 10.100.0.6:0 | 10.100.0.1 | 10.100.0.6, 100.100.1.2, 100.100.1.17, 100.100.1.42 |
| 2 | DOp | 10.100.0.9 | | 10.100.0.9:0 | 10.100.0.1 | 10.100.0.9, 100.100.1.6 |
| 3 | DOp | 10.100.0.5 | | 10.100.0.5:0 | 10.100.0.1 | 10.100.0.5, 100.100.1.9, 100.100.1.34 |
| 4 | DOp | 10.100.0.2 | | 10.100.0.2:0 | 10.100.0.1 | 10.100.0.2, 100.100.1.14, 100.100.1.18, 100.100.1.21, 100.100.1.25, 100.100.1.30 |
| 5 | DOp | 10.100.0.11 | | 10.100.0.11:0 | 10.100.0.1 | 10.100.0.11, 100.100.1.65 |
| 6 | DOp | 10.100.0.12 | | 10.100.0.12:0 | 10.100.0.1 | 10.100.0.12, 100.100.1.61 |

R2



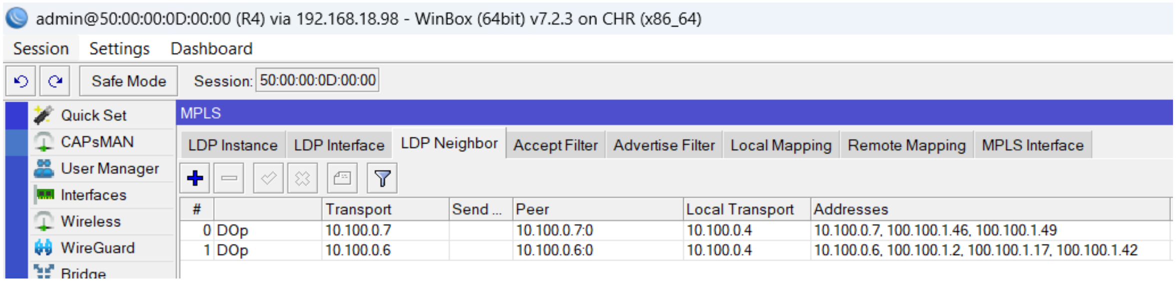

| # | | Transport | Send ... | Peer | Local Transport | Addresses |
|---|---|---|---|---|---|---|
| 0 | DOp | 10.100.0.6 | | 10.100.0.6:0 | 10.100.0.2 | 10.100.0.6, 100.100.1.2, 100.100.1.17, 100.100.1.42 |
| 1 | DOp | 10.100.0.3 | | 10.100.0.3:0 | 10.100.0.2 | 10.100.0.3, 100.100.1.10, 100.100.1.13 |
| 2 | DOp | 10.100.0.10 | | 10.100.0.10:0 | 10.100.0.2 | 10.100.0.10, 100.100.1.22 |
| 3 | DOp | 10.100.0.13 | | 10.100.0.13:0 | 10.100.0.2 | 10.100.0.13, 100.100.1.26 |
| 4 | DO | 10.100.0.1 | | 10.100.0.1:0 | 10.100.0.2 | 10.100.0.1, 100.100.1.1, 100.100.1.5, 100.100.1.29, 100.100.1.33, 100.100.1.54, 100.100.1.62, 100.100.1.66, 100.100.1.253, 192.168.10.254, 192.168.18.98 |

R3



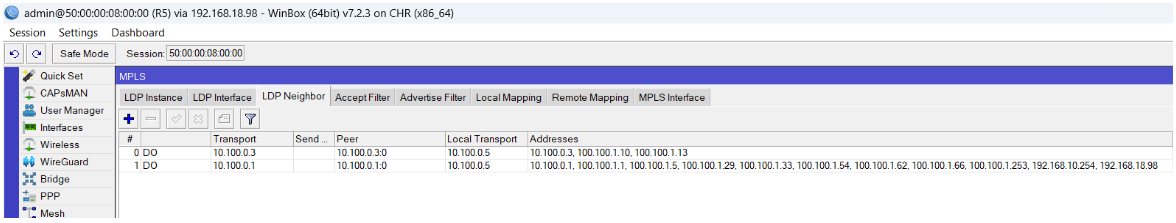

| # | | Transport | Send ... | Peer | Local Transport | Addresses |
|---|---|---|---|---|---|---|
| 0 | DO | 10.100.0.2 | | 10.100.0.2:0 | 10.100.0.3 | 10.100.0.2, 100.100.1.14, 100.100.1.18, 100.100.1.21, 100.100.1.25, 100.100.1.30 |
| 1 | DOp | 10.100.0.5 | | 10.100.0.5:0 | 10.100.0.3 | 10.100.0.5, 100.100.1.9, 100.100.1.34 |

R4



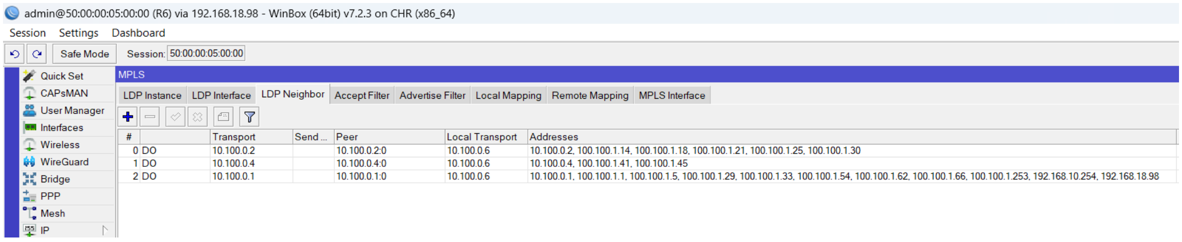

| # | | Transport | Send ... | Peer | Local Transport | Addresses |
|---|---|---|---|---|---|---|
| 0 | DOp | 10.100.0.7 | | 10.100.0.7:0 | 10.100.0.4 | 10.100.0.7, 100.100.1.46, 100.100.1.49 |
| 1 | DOp | 10.100.0.6 | | 10.100.0.6:0 | 10.100.0.4 | 10.100.0.6, 100.100.1.2, 100.100.1.17, 100.100.1.42 |

R5



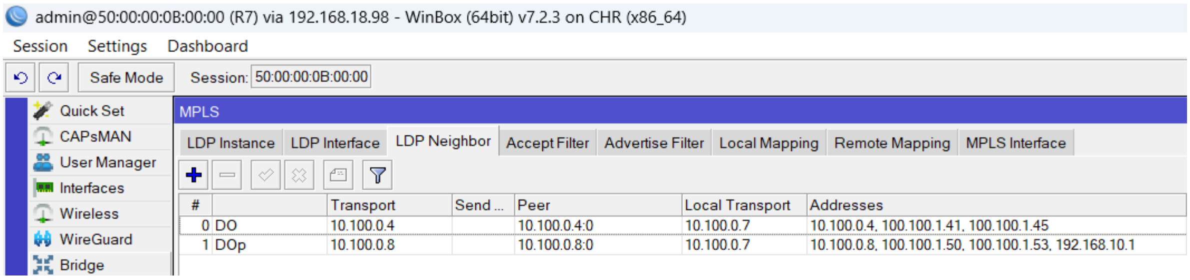

| # | | Transport | Send ... | Peer | Local Transport | Addresses |
|---|---|---|---|---|---|---|
| 0 | DO | 10.100.0.3 | | 10.100.0.3:0 | 10.100.0.5 | 10.100.0.3, 100.100.1.10, 100.100.1.13 |
| 1 | DO | 10.100.0.1 | | 10.100.0.1:0 | 10.100.0.5 | 10.100.0.1, 100.100.1.1, 100.100.1.5, 100.100.1.29, 100.100.1.33, 100.100.1.54, 100.100.1.62, 100.100.1.66, 100.100.1.253, 192.168.10.254, 192.168.18.98 |

R6



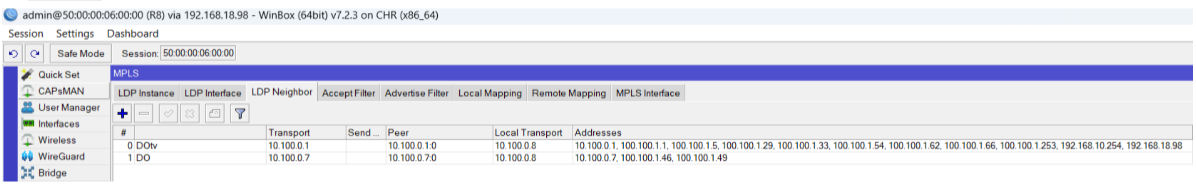

| # | | Transport | Send ... | Peer | Local Transport | Addresses |
|---|---|---|---|---|---|---|
| 0 | DO | 10.100.0.2 | | 10.100.0.2:0 | 10.100.0.6 | 10.100.0.2, 100.100.1.14, 100.100.1.18, 100.100.1.21, 100.100.1.25, 100.100.1.30 |
| 1 | DO | 10.100.0.4 | | 10.100.0.4:0 | 10.100.0.6 | 10.100.0.4, 100.100.1.41, 100.100.1.45 |
| 2 | DO | 10.100.0.1 | | 10.100.0.1:0 | 10.100.0.6 | 10.100.0.1, 100.100.1.1, 100.100.1.5, 100.100.1.29, 100.100.1.33, 100.100.1.54, 100.100.1.62, 100.100.1.66, 100.100.1.253, 192.168.10.254, 192.168.18.98 |

R7



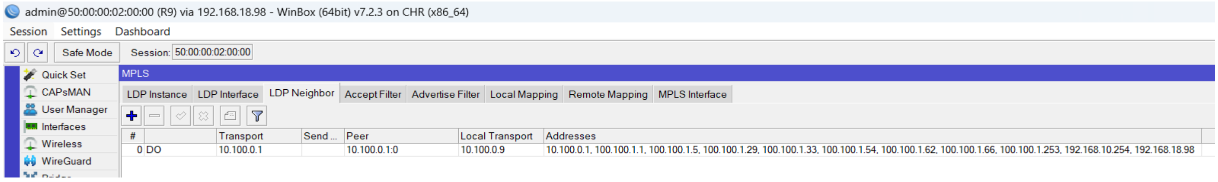

| # | | Transport | Send ... | Peer | Local Transport | Addresses |
|---|---|---|---|---|---|---|
| 0 | DO | 10.100.0.4 | | 10.100.0.4:0 | 10.100.0.7 | 10.100.0.4, 100.100.1.41, 100.100.1.45 |
| 1 | DOp | 10.100.0.8 | | 10.100.0.8:0 | 10.100.0.7 | 10.100.0.8, 100.100.1.50, 100.100.1.53, 192.168.10.1 |

R8

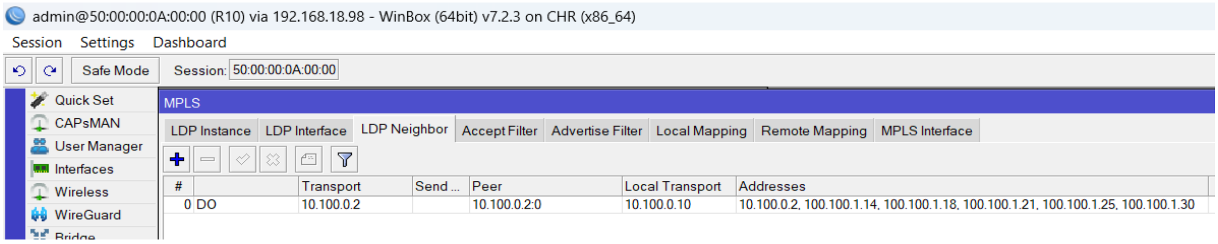



| # | | Transport | Send ... | Peer | Local Transport | Addresses |
|---|---|---|---|---|---|---|
| 0 | DOtv | 10.100.0.1 | | 10.100.0.1:0 | 10.100.0.8 | 10.100.0.1, 100.100.1.1, 100.100.1.5, 100.100.1.29, 100.100.1.33, 100.100.1.54, 100.100.1.62, 100.100.1.66, 100.100.1.253, 192.168.10.254, 192.168.18.98 |
| 1 | DO | 10.100.0.7 | | 10.100.0.7:0 | 10.100.0.8 | 10.100.0.7, 100.100.1.46, 100.100.1.49 |

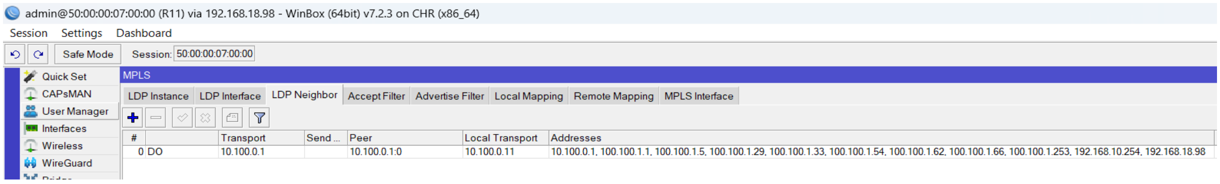
R9



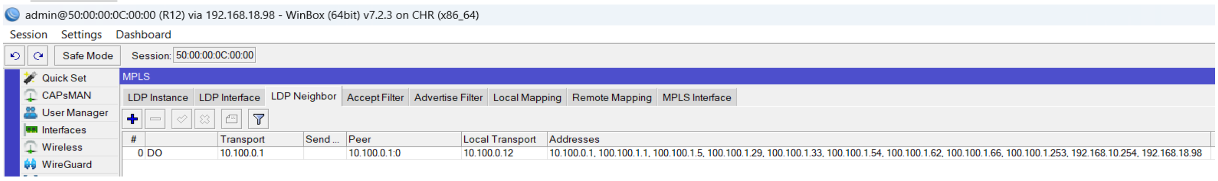

R10



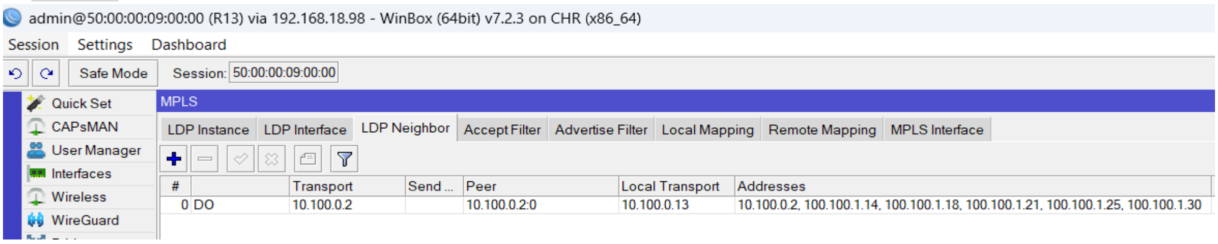

R11




R12




R13




- **Túnel VPLS**

  Para a conexão L2 entre R1 e R8, foi criada uma interface VPLS em cada lado. Como *Remote Peer*, foram utilizados os endereços de Loopback dos roteadores, garantindo a comunicação L2 independentemente de quaisquer eventualidades nos enlaces físicos. Nos dois lados, a interface VPLS foi adicionada a uma Bridge juntamente com a interface física ligada ao VPC (ether10-VPC). No lado do R8, a Bridge fornece endereçamento IP aos dispositivos pertencentes ao domínio L2 via DHCP Server. No lado do R1, a Bridge recebe essas informações via DHCP Client, permitindo que a VPC-R1 consiga endereçamento.

➢ Interfaces VPLS R1 e R8

```
[admin@R1] > interface/vpls/print
Flags: R - RUNNING
Columns: NAME, PEER, VPLS-ID
#   NAME     PEER       VPLS-ID
0 R vpls-R8  10.100.0.8  1:1
```

```
[admin@R8] > interface/vpls/print
Flags: R - RUNNING
Columns: NAME, PEER, VPLS-ID
#   NAME     PEER       VPLS-ID
0 R vpls-R1  10.100.0.1  1:1
```

➢ Bridge R1 e R8

```
[admin@R1] > interface/bridge/port/print
Flags: D - DYNAMIC
Columns: INTERFACE, BRIDGE, HW, PVID, PRIORITY, PATH-COST, INTERNAL-PATH-COST, H
ORIZON
#   INTERFACE   BRIDGE   HW   PVID  PRIORITY  PATH-COST  IN  HORIZON
0   ether10-VPC  VPLS-R8  yes    1  0x80         10  10  none
1 D vpls-R8     VPLS-R8            1  0x80         10  10  none
```

```
[admin@R8] > interface/bridge/port/print
Flags: D - DYNAMIC
Columns: INTERFACE, BRIDGE, HW, PVID, PRIORITY, PATH-COST, INTERNAL-PATH-COST,
HORIZON
#   INTERFACE   BRIDGE   HW   PVID  PRIORITY  PATH-COST  IN  HORIZON
0   ether10-VPC  VPLS-R1  yes    1  0x80         10  10  none
1 D vpls-R1     VPLS-R1            1  0x80         10  10  none
```

➢ DHCP R1 e R8

```
[admin@R1] > ip dhcp-client/print
Columns: INTERFACE, USE-PEER-DNS, ADD-DEFAULT-ROUTE, STATUS, ADDRESS
# INTERFACE  USE-PEER-DNS  ADD-DEFAULT-ROUTE  STATUS  ADDRESS
0 ether1     yes          yes             bound   192.168.18.98/24
1 VPLS-R8    no           no              bound   192.168.10.254/24
```

```
[admin@R8] > ip dhcp-server/print
Columns: NAME, INTERFACE, ADDRESS-POOL, LEASE-TIME
# NAME   INTERFACE  ADDRESS-POOL  LEASE-TIME
0 dhcp1  VPLS-R1    dhcp_pool0    10m
```

➢ Tabela de Macs R1 e R8

```
[admin@R1] > interface/bridge/host/print
Flags: D - DYNAMIC; L - LOCAL
Columns: MAC-ADDRESS, ON-INTERFACE, BRIDGE, AGE
#    MAC-ADDRESS       ON-INTERFACE  BRIDGE    AGE
0 DL 0E:9C:FB:1B:24:F6  loopback      loopback
1 D  00:50:79:66:68:0E  vpls-R8       VPLS-R8   3m54s
2 D  00:50:79:66:68:0F  ether10-VPC   VPLS-R8   3m46s
3 DL 02:77:DC:C2:17:BB  vpls-R8       VPLS-R8
4 D  02:7F:6B:7B:28:05  vpls-R8       VPLS-R8   11s
5 DL 50:00:00:01:00:09  ether10-VPC   VPLS-R8
6 D  50:00:00:06:00:09  vpls-R8       VPLS-R8   1m12s
```

```
[admin@R8] > interface/bridge/host/print
Flags: D - DYNAMIC; L - LOCAL
Columns: MAC-ADDRESS, ON-INTERFACE, BRIDGE, AGE
#    MAC-ADDRESS       ON-INTERFACE  BRIDGE    AGE
0 DL 86:7D:EE:34:DD:4F  loopback      loopback
1 D  00:50:79:66:68:0E  ether10-VPC   VPLS-R1   4m3s
2 D  00:50:79:66:68:0F  vpls-R1       VPLS-R1   3m55s
3 D  02:77:DC:C2:17:BB  vpls-R1       VPLS-R1   2s
4 DL 02:7F:6B:7B:28:05  vpls-R1       VPLS-R1
5 D  50:00:00:01:00:09  vpls-R1       VPLS-R1   1m21s
6 DL 50:00:00:06:00:09  ether10-VPC   VPLS-R1
```

- Testes de comunicação entre VPC-R1 e VPC-R8

VPC-R1

```
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> dhcp
DORA IP 192.168.10.253/24 GW 192.168.10.1

VPCS> ping 192.168.10.252

84 bytes from 192.168.10.252 icmp_seq=1 ttl=64 time=2.616 ms
84 bytes from 192.168.10.252 icmp_seq=2 ttl=64 time=2.885 ms
84 bytes from 192.168.10.252 icmp_seq=3 ttl=64 time=3.033 ms
84 bytes from 192.168.10.252 icmp_seq=4 ttl=64 time=3.313 ms
84 bytes from 192.168.10.252 icmp_seq=5 ttl=64 time=2.877 ms

VPCS>
```

VPC-R8

```
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> dhcp
DORA IP 192.168.10.252/24 GW 192.168.10.1

VPCS> ping 192.168.10.253

84 bytes from 192.168.10.253 icmp_seq=1 ttl=64 time=2.348 ms
84 bytes from 192.168.10.253 icmp_seq=2 ttl=64 time=2.607 ms
84 bytes from 192.168.10.253 icmp_seq=3 ttl=64 time=3.105 ms
84 bytes from 192.168.10.253 icmp_seq=4 ttl=64 time=2.568 ms
84 bytes from 192.168.10.253 icmp_seq=5 ttl=64 time=2.851 ms

VPCS>
```

# 5.Servidor de aplicação, Nginx

Simulando uma aplicação, seguindo os requisitos, um servidor Linux com Nginx instalado foi adicionado na topologia, estando acessível por toda a rede. O servidor está conectado diretamente no R1, pela interface física ether9, sendo adicionada no OSPF para a distribuição do acesso aos demais equipamentos do Backbone.

- Compartilhamento de rota padrão via OSPF

O OSPF já compartilha uma rota padrão, o que já possibilitaria a conectividade dos equipamentos com o servidor, mas, por boa prática, a rede do servidor foi adicionada no roteamento dinâmico em modo passivo, juntamente com a Loopback.

```
[admin@R1] > routing/ospf/instance/print
Flags: X - disabled, I - inactive
 0   name="ospf-instance-1" version=2 vrf=main router-id=10.100.0.1
     originate-default=always redistribute=""
```

```
[admin@R1] > routing/ospf/interface-template/print
Flags: X - disabled, I - inactive
 0   area=ospf-area-1 interfaces=loopback,ether9-Nginx instance-id=0
     type=broadcast retransmit-interval=5s transmit-delay=1s
     hello-interval=10s dead-interval=40s priority=128 cost=1 passive
     auth=md5 auth-key="@Dm1122ti#"
```

- Teste de comunicação do R7 com o Nginx

```
[admin@R7] > /tool fetch url="http://100.100.1.254" dst-path=index.html
      status: finished
  downloaded: 0KiBC-z pause]
       total: 0KiB
    duration: 1s


[admin@R7] > file/print where name~"index.html"
Columns: NAME, TYPE, SIZE, CREATION-TIME
#  NAME       TYPE       SIZE  CREATION-TIME
0  index.html  .html file   612  sep/02/2026 16:23:58
```

- Página HTML Nginx


## 6.Testes específicos solicitados R3 <-> R7

- PING

```
R3
[admin@R3] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS           NETWORK       INTERFACE
;;; LOOPBACK
0 10.100.0.3/32     10.100.0.3    loopback
;;; PARA-R5
1 100.100.1.10/30   100.100.1.8   ether3-R5
;;; PARA-R2
2 100.100.1.13/30   100.100.1.12  ether2-R2
[admin@R3] > ping 10.100.0.7
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.100.0.7                                 56  61 2ms255us
    1 10.100.0.7                                 56  61 2ms241us
    2 10.100.0.7                                 56  61 2ms576us
    3 10.100.0.7                                 56  61 2ms516us
    4 10.100.0.7                                 56  61 2ms241us
    5 10.100.0.7                                 56  61 2ms427us
    6 10.100.0.7                                 56  61 2ms903us
    7 10.100.0.7                                 56  61 2ms345us
    8 10.100.0.7                                 56  61 2ms413us
    9 10.100.0.7                                 56  61 2ms701us
   10 10.100.0.7                                 56  61 3ms571us
   11 10.100.0.7                                 56  61 2ms139us
    sent=12 received=12 packet-loss=0% min-rtt=2ms139us avg-rtt=2ms527us
```

```
R7
[admin@R7] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS           NETWORK       INTERFACE
;;; LOOPBACK
0 10.100.0.7/32     10.100.0.7    loopback
;;; PARA-R4
1 100.100.1.46/30   100.100.1.44  ether2-R4
;;; PARA-R8
2 100.100.1.49/30   100.100.1.48  ether1-R8
[admin@R7] > ping 10.100.0.3
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.100.0.3                                 56  61 2ms534us
    1 10.100.0.3                                 56  61 2ms161us
    2 10.100.0.3                                 56  61 2ms427us
    3 10.100.0.3                                 56  61 2ms654us
    4 10.100.0.3                                 56  61 2ms495us
    5 10.100.0.3                                 56  61 2ms184us
    6 10.100.0.3                                 56  61 2ms739us
    7 10.100.0.3                                 56  61 2ms247us
    8 10.100.0.3                                 56  61 3ms371us
    9 10.100.0.3                                 56  61 2ms447us
   10 10.100.0.3                                 56  61 2ms535us
    sent=11 received=11 packet-loss=0% min-rtt=2ms161us avg-rtt=2ms526us
  max-rtt=3ms371us
```

➢ Traceroute

```
[admin@R3] > tool/traceroute 10.100.0.7
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
#  ADDRESS        LOSS   SENT  LAST    AVG  BEST  WORST  STD-DEV  STATUS
1  100.100.1.14   0%       23  2.7ms   2.7  1.7   7.4    1.1      <MPLS:L=22,E=0>
2  100.100.1.17   0%       23  2.1ms   2.6  1.7   6.9    1.1      <MPLS:L=24,E=0>
3  100.100.1.41   0%       23  2.1ms   2.2  1.6   3.4    0.5      <MPLS:L=21,E=0>
4  10.100.0.7     0%       23  2ms     2.3  1.6   7      1.2
-- [Q quit|D dump|C-z pause]
```

```
[admin@R7] > tool/traceroute 10.100.0.3
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
#  ADDRESS        LOSS   SENT  LAST    AVG  BEST  WORST  STD-DEV  STATUS
1  100.100.1.45   0%        2  2.6ms   3.1  2.6   3.6    0.5      <MPLS:L=18,E=0>
2  100.100.1.42   0%        2  2ms     2.4  2     2.7    0.4      <MPLS:L=28,E=0>
3  100.100.1.18   0%        2  2.1ms   2.3  2.1   2.5    0.2      <MPLS:L=24,E=0>
4  10.100.0.3     0%        2  2.1ms   2.3  2.1   2.4    0.2
-- [Q quit|D dump|C-z pause]
```

# 7.Testes de redundância

O teste envolve uma comunicação ICMP do Router R13 para o R1. Como resultado, nas imagens abaixo, será mostrado o caminho inicial usado até o endereço do R1, o impacto após a queda do caminho principal e a restauração da comunicação por meio de outro caminho.

R13

```
n
tdown
sep/02/2026 02:29:49 system,error,critical router was rebooted without proper sh
n
tdown
sep/02/2026 03:38:07 system,error,critical login failure for user admina via loc
a
l
sep/02/2026 12:44:15 system,error,critical router was rebooted without proper sh
n
tdown
sep/03/2026 18:17:55 system,error,critical router was rebooted without proper sh
n
tdown
sep/03/2026 20:02:15 system,error,critical login failure for user amdin via loca
l


[admin@R13] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
#  ADDRESS        LOSS  SENT  LAST    AVG  BEST  WORST  STD-DEV  STATUS
1  100.100.1.25   0%     173  1.1ms   1.4  0.9   6.1    0.6      <MPLS:L=21,E=0>
2  10.100.0.1     0%     173  0.9ms   1.1  0.7   3.9    0.4
-- [Q quit|D dump|C-z pause]
```

- **Interface ether4 Down (Caminho Principal)**

```
R1
Columns: NAME, TYPE, ACTUAL-MTU, L2MTU, MAC-ADDRESS
 #    NAME          TYPE    ACTUAL-MTU  L2MTU  MAC-ADDRESS
;;; NET
 0 R  ether1        ether         1500         50:00:00:01:00:00
;;; R6
 1 R  ether2-R6     ether         9000         50:00:00:01:00:01
;;; R9
 2 R  ether3-R9     ether         1500         50:00:00:01:00:02
;;; R5
 3 R  ether4-R5     ether         1500         50:00:00:01:00:03
;;; R2
 4 R  ether5-R2     ether         9000         50:00:00:01:00:04
;;; R12
 5 R  ether6-R12    ether         1500         50:00:00:01:00:05
;;; R11
 6 R  ether7-R11    ether         1500         50:00:00:01:00:06
;;; R8
 7 R  ether8-R8     ether         9000         50:00:00:01:00:07
;;; Nginx
 8 R  ether9-Nginx  ether         1500         50:00:00:01:00:08
;;; LAN
 9 RS ether10-VPC   ether         1500         50:00:00:01:00:09
10 R  VPLS-R8       bridge        1500  65535  50:00:00:01:00:09

[admin@R1] > interface/disable number=4
```

- **Perda de comunicação com R1**

```
R13
u
tdown
sep/02/2026 02:29:49 system,error,critical router was rebooted without proper sh
u
tdown
sep/02/2026 03:38:07 system,error,critical login failure for user admina via loc
a
l
sep/02/2026 12:44:15 system,error,critical router was rebooted without proper sh
u
tdown
sep/03/2026 18:17:55 system,error,critical router was rebooted without proper sh
u
tdown
sep/03/2026 20:02:15 system,error,critical login failure for user amdin via loca
l


[admin@R13] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
#  ADDRESS       LOSS  SENT  LAST     AVG  BEST  WORST  STD-DEV  STATUS        >
1  100.100.1.25  0.2%   965  timeout  1.4  0.8   23.6   0.9      <MPLS:L=21,E=>
2  10.100.0.1    0.1%   965  timeout  1.1  0.6   18.3   0.7                    >
-- [Q quit|D dump|C-z pause|right]
```

- **Retomada da conexão por outro caminho**

```
R13
sep/02/2026 12:44:15 system,error,critical router was rebooted without proper sh
u
tdown
sep/03/2026 18:17:55 system,error,critical router was rebooted without proper sh
u
tdown
sep/03/2026 20:02:15 system,error,critical login failure for user amdin via loca
l


[admin@R13] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
#  ADDRESS        LOSS  SENT  LAST   AVG  BEST  WORST  STD-DEV  STATUS
1  100.100.1.25   0.6%   996  1.8ms  1.4  0.8   23.6   0.9      <MPLS:L=21,E=0>
2  10.100.0.1     0.5%   996  1.6ms  1.1  0.6   18.3   0.7      <MPLS:L=35,E=0>
   100.100.1.17

[admin@R13] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
#  ADDRESS        LOSS  SENT  LAST   AVG  BEST  WORST  STD-DEV  STATUS
1  100.100.1.25   0%       6  1.8ms  2    1.7   2.5    0.3      <MPLS:L=21,E=0>
2  100.100.1.17   0%       6  1.6ms  1.6  1.3   1.8    0.2      <MPLS:L=35,E=0>
3  10.100.0.1     0%       6  1.5ms  1.5  1.3   1.9    0.2
-- [Q quit|D dump|C-z pause]
```

- **Endereços relacionados**

```
R2
tdown
sep/02/2026 12:44:22 system,error,critical router was rebooted without proper sh
u
tdown
sep/03/2026 18:18:04 system,error,critical router was rebooted without proper sh
u
tdown

[admin@R2] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS            NETWORK       INTERFACE
;;; LOOPBACK
0 10.100.0.2/32      10.100.0.2    loopback
;;; PARA-R3
1 100.100.1.14/30    100.100.1.12  ether2-R3
;;; PARA-R6
2 100.100.1.18/30    100.100.1.16  ether1-R6
;;; PARA-R10
3 100.100.1.21/30    100.100.1.20  ether3-R10
;;; PARA-R13
4 100.100.1.25/30    100.100.1.24  ether4-R13
;;; PARA-R1
5 100.100.1.30/30    100.100.1.28  ether5-R1
[admin@R2] >
```

```
[admin@R6] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS           NETWORK       INTERFACE
;;; LOOPBACK
0 10.100.0.6/32     10.100.0.6    loopback
;;; PARA-R1
1 100.100.1.2/30    100.100.1.0   ether2-R1
;;; PARA-R2
2 100.100.1.17/30   100.100.1.16  ether1-R2
;;; PARA-R4
3 100.100.1.42/30   100.100.1.40  ether3-R4
[admin@R6] >
```

- **Validação da preferência de rota R8<->R1**

```
[admin@R1] > tool/traceroute 10.100.0.8
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
#  ADDRESS        LOSS  SENT  LAST    AVG  BEST  WORST  STD-DEV  STATUS
1  100.100.1.2    0%      48  2ms     3    2     12.7   1.8      <MPLS:L=27,E=0>
2  100.100.1.41   0%      48  1.6ms   2.5  1.6   13.2   1.8      <MPLS:L=22,E=0>
3  100.100.1.46   0%      48  1.6ms   2.4  1.6   20.2   2.7      <MPLS:L=17,E=0>
4  10.100.0.8     0%      48  1.8ms   2.3  1.5   10.7   1.4
-- [Q quit|D dump|C-z pause]
```

```
[admin@R8] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
#  ADDRESS        LOSS  SENT  LAST    AVG  BEST  WORST  STD-DEV  STATUS
1  100.100.1.49   0%      46  3ms     2.6  1.8   6.3    0.7      <MPLS:L=20,E=0>
2  100.100.1.45   0%      46  1.9ms   2.4  1.4   10.6   1.4      <MPLS:L=16,E=0>
3  100.100.1.42   0%      46  1.9ms   2.4  1.5   16.4   2.1      <MPLS:L=35,E=0>
4  10.100.0.1     0%      46  2.1ms   2.3  1.5   14     1.8
-- [Q quit|D dump|C-z pause]
```

- **Queda de interface com menor custo**

```
[admin@R7] > interface/p
Flags: R - RUNNING
Columns: NAME, TYPE, ACTUAL-MTU, L2MTU, MAC-ADDRESS
 #   NAME        TYPE    ACTUAL-MTU  L2MTU  MAC-ADDRESS
;;; R8
 0 R ether1-R8   ether         9000         50:00:00:0B:00:00
;;; R4
 1 R ether2-R4   ether         9000         50:00:00:0B:00:01
 2 R ether3      ether         1500         50:00:00:0B:00:02
 3 R ether4      ether         1500         50:00:00:0B:00:03
 4 R ether5      ether         1500         50:00:00:0B:00:04
 5 R ether6      ether         1500         50:00:00:0B:00:05
 6 R ether7      ether         1500         50:00:00:0B:00:06
 7 R ether8      ether         1500         50:00:00:0B:00:07
 8 R ether9      ether         1500         50:00:00:0B:00:08
 9 R ether10     ether         1500         50:00:00:0B:00:09
10 R loopback    bridge        1500  65535  66:02:67:07:AC:A2
[admin@R7] > interface/disable number=0
```

- **Perda de comunicação**

```
[admin@R8] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
#  ADDRESS        LOSS  SENT  LAST     AVG  BEST  WORST  STD-DEV  STATUS        >
1  100.100.1.49   0.8%   387  timeout  2.8  1.8   13.4   1.3      <MPLS:L=20,E=>
2  100.100.1.45   0.8%   387  timeout  2.3  1.4   10.6   0.9      <MPLS:L=16,E=>
3  100.100.1.42   0.5%   387  timeout  2.2  1.3   16.4   1        <MPLS:L=35,E=>
4  10.100.0.1     0.5%   386  timeout  2.2  1.3   14     1.2                    >
-- [Q quit|D dump|C-z pause|right]
```

```
[admin@R1] > tool/traceroute 10.100.0.8
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
#  ADDRESS        LOSS  SENT  LAST     AVG  BEST  WORST  STD-DEV  STATUS        >
1  100.100.1.2    1.1%   376  timeout  2.7  1.8   12.7   1.1      <MPLS:L=27,E=>
2  100.100.1.41   0.8%   376  timeout  2.3  1.5   13.2   1.1      <MPLS:L=22,E=>
3  100.100.1.46   0.8%   375  timeout  2.2  1.4   20.2   1.2      <MPLS:L=17,E=>
4  10.100.0.8     0.8%   375  timeout  2.1  1.4   12.1   0.9                    >
-- [Q quit|D dump|C-z pause|right]
```

- **Restauração pelo caminho de maior custo (1000)**
  R8 possui uma ligação direta com o R1, porém, no OSPF foi definido o custo de 1000 para o enlace.

```
[admin@R8] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV
#  ADDRESS      LOSS  SENT  LAST   AVG  BEST  WORST  STD-DEV
1  10.100.0.1   0%       8  0.7ms  0.7  0.5   1.1    0.2
-- [Q quit|D dump|C-z pause]
```

```
[admin@R1] > tool/traceroute 10.100.0.8
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV
#  ADDRESS      LOSS  SENT  LAST   AVG  BEST  WORST  STD-DEV
1  10.100.0.8   0%      18  0.6ms  0.8  0.5   1.6    0.3
-- [Q quit|D dump|C-z pause]
```

- **Endereços relacionados**

```
[admin@R7] > ip address/print
Flags: I, D - DYNAMIC
Columns: ADDRESS, NETWORK, INTERFACE
#   ADDRESS          NETWORK       INTERFACE
;;; LOOPBACK
0   10.100.0.7/32    10.100.0.7    loopback
;;; PARA-R4
1   100.100.1.46/30  100.100.1.44  ether2-R4
;;; PARA-R8
2 I 100.100.1.49/30  100.100.1.48  ether1-R8
[admin@R7] >
```

**A interface em azul foi desabilitada para o teste**

```
[admin@R6] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS           NETWORK       INTERFACE
;;; LOOPBACK
0 10.100.0.6/32     10.100.0.6    loopback
;;; PARA-R1
1 100.100.1.2/30    100.100.1.0   ether2-R1
;;; PARA-R2
2 100.100.1.17/30   100.100.1.16  ether1-R2
;;; PARA-R4
3 100.100.1.42/30   100.100.1.40  ether3-R4
[admin@R6] >
```

- **Caminho principal restabelecido**

Interface habilitada

```
[admin@R7] > interface/enable number=0
[admin@R7] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS           NETWORK       INTERFACE
;;; LOOPBACK
0 10.100.0.7/32     10.100.0.7    loopback
;;; PARA-R4
1 100.100.1.46/30   100.100.1.44  ether2-R4
;;; PARA-R8
2 100.100.1.49/30   100.100.1.48  ether1-R8
[admin@R7] >
```

```
[admin@R1] > tool/traceroute 10.100.0.8
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
#  ADDRESS        LOSS  SENT  LAST   AVG  BEST  WORST  STD-DEV  STATUS
1  100.100.1.2    0%      86  2.6ms  3.6  2.1   23.3   3        <MPLS:L=27,E=0>
2  100.100.1.41   0%      86  2ms    2.8  1.6   11.8   1.6      <MPLS:L=22,E=0>
3  100.100.1.46   0%      86  2.2ms  2.7  1.6   10.8   1.5      <MPLS:L=17,E=0>
4  10.100.0.8     0%      86  3.9ms  3    1.6   14.5   2.1
-- [Q quit|D dump|C-z pause]
```

Caminho Principal Restaurado

```
[admin@R8] > tool/traceroute 10.100.0.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
#  ADDRESS        LOSS  SENT  LAST   AVG  BEST  WORST  STD-DEV  STATUS
1  100.100.1.49   0%     139  6.9ms  2.9  1.4   9.1    1.3      <MPLS:L=20,E=0>
2  100.100.1.45   0%     139  2ms    2.6  1.4   19.9   1.9      <MPLS:L=16,E=0>
3  100.100.1.42   0%     139  2.1ms  2.4  1.4   8.7    1.2      <MPLS:L=35,E=0>
4  10.100.0.1     0%     139  1.9ms  2.4  1.4   13.2   1.3
-- [Q quit|D dump|C-z pause]
```

Caminho Principal Restaurado

# 8. Troubleshooting

- ## State Exchange

Na verificação final das configurações dos Routers, foi identificado que o state na vizinhança do R1 com R2 estava travado em “Exchange”, após verificações, foi visto uma divergência nos MTUs das interfaces do enlace. Feito a correção, os Routers ficaram com state “Full” novamente.

```
[admin@R1] > routing/ospf/neighbor/print where state~"Exchange"
Flags: V - virtual; D - dynamic
 3  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.30
      router-id=10.100.0.2 state="Exchange" state-changes=4 ls-retransmits=14
      timeout=33s
```

```
[admin@R1] > interface/print where name~"ether5"
Flags: R - RUNNING
Columns: NAME, TYPE, ACTUAL-MTU, MAC-ADDRESS
#   NAME    TYPE   ACTUAL-MTU  MAC-ADDRESS
;;; R2
0 R ether5  ether        9000  50:00:00:01:00:04
[admin@R1] >
```

```
[admin@R2] > interface/print where name~"ether5"
Flags: R - RUNNING
Columns: NAME, TYPE, ACTUAL-MTU, MAC-ADDRESS
#   NAME    TYPE   ACTUAL-MTU  MAC-ADDRESS
;;; R1
0 R ether5  ether        1500  50:00:00:03:00:04
[admin@R2] >
```

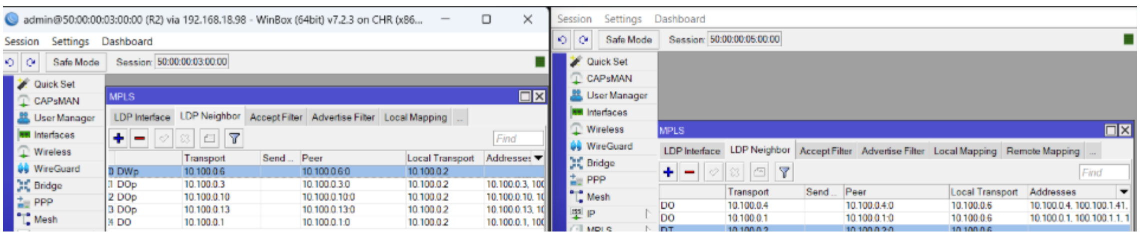
```
[admin@R1] > routing/ospf/neighbor/print where address ~"100.100.1.30"
Flags: V - virtual; D - dynamic
 3  D instance=ospf-instance-1 area=ospf-area-1 address=100.100.1.30
      router-id=10.100.0.2 state="Full" state-changes=5 adjacency=1m15s
      timeout=40s
[admin@R1] >
```

- ## Falha no MPLS R2 com R6

Ainda na validação de configuração dos Routers, foi feito um teste do R2 para o R8 e foi percebido que o MPLS não atuava ao passar pelo R6. A imagem mostra isso claramente com o status nulo, sem a Label MPLS

```
admin@R2] > tool/traceroute 10.100.0.8
olumns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV, STATUS
  ADDRESS        LOSS  SENT  LAST    AVG  BEST  WORST  STD-DEV  STATUS
  100.100.1.17   0%     342  0.9ms   1    0.4   28.9   2.3
  100.100.1.41   0%     342  2.5ms   2.9  1.4   89.4   5        <MPLS:L=21,E=0>
  100.100.1.46   0%     342  2.7ms   2.6  1.4   49.4   3.2      <MPLS:L=19,E=0>
  10.100.0.8     0%     342  2.4ms   2.3  1.4   12     1.1
- [Q quit|D dump|C-z pause]
```

Ao visitar as configurações do MPLS no R2 e no R6, foi visto que o campo ADDRESSES, em ambos os Routers, estava sem informação.



Após algumas verificações, chegou o momento de checar os endereços de ambos os Routers. Foi identificado que o endereço da rede de Loopback do R6 estava com o endereço da interface de Loopback do R2, impedindo a comunicação.

```
[admin@R2] > /ip/address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS           NETWORK       INTERFACE
;;; LOOPBACK
0 10.100.0.2/32     10.100.0.2    loopback
;;; PARA-R3
1 100.100.1.14/30   100.100.1.12  ether2
;;; PARA-R6
2 100.100.1.18/30   100.100.1.16  ether1
;;; PARA-R10
3 100.100.1.21/30   100.100.1.20  ether3
;;; PARA-R13
4 100.100.1.25/30   100.100.1.24  ether4
;;; PARA-R1
5 100.100.1.30/30   100.100.1.28  ether5
[admin@R2] >
```

```
 2  D instance=ospf-instance-1 area=ospf-area-1 addr
      router-id=10.100.0.4 state="Full" state-chang
      timeout=39s
[admin@R6] > /ip/address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS           NETWORK       INTERFACE
;;; LOOPBACK
0 10.100.0.6/32     10.100.0.2    loopback
;;; PARA-R1
1 100.100.1.2/30    100.100.1.0   ether2
;;; PARA-R2
2 100.100.1.17/30   100.100.1.16  ether1
;;; PARA-R4
3 100.100.1.42/30   100.100.1.40  ether3
[admin@R6] >
```

Ao identificar o endereço da Rede configurado na interface da Loopback, o RouterOs criou uma rota diretamente conectada com a distância 0. Como por padrão a distância das rotas OSPF é 110, o router escolhia a rota com o menor custo e enviava pacotes para ele mesmo. O segundo efeito, era a falta de comunicação dos outros Routers com o R6 através do endereço de Loopback.

```
5 100.100.1.30/30  100.100.1.28  ether5
[admin@R2] > /ip/route/print where dst-address~"10.100.0.2"
Flags: D - DYNAMIC; A - ACTIVE; c, y - COPY
Columns: DST-ADDRESS, GATEWAY, DISTANCE
    DST-ADDRESS    GATEWAY    DISTANCE
DAc 10.100.0.2/32  loopback          0
[admin@R2] >
```

```
[admin@R6] > /ip/route/print where dst-address~"10.100.0.2"
Flags: D - DYNAMIC; A - ACTIVE; c, o, y - COPY
Columns: DST-ADDRESS, GATEWAY, DISTANCE
    DST-ADDRESS    GATEWAY                DISTANCE
D o 10.100.0.2/32  100.100.1.18%ether1         110
DAc 10.100.0.2/32  loopback                      0
[admin@R6] >
```

## 9.Configurações adicionais no Firewall do R8

A fim de permitir apenas as conexões necessárias para a Lan da VPLS, foi configurado regras no firewall do R8 (Servidor DHCP) para impedir a comunicação com as Loopbacks e com os endereços das interfaces físicas dos Routers. Perceba que uma das regras possui uma exceção, permitindo o tráfego usando qualquer protocolo para o endereço do servidor de aplicação Nginx.

```
[admin@R8] > ip firewall/address-list/print
Columns: LIST, ADDRESS, CREATION-TIME
# LIST  ADDRESS         CREATION-TIME
0 IPS   10.100.0.0/24   aug/31/2026 01:48:22
1 IPS   100.100.1.0/24  aug/31/2026 02:14:10
[admin@R8] > ip firewall/filter/print
Flags: X - disabled, I - invalid; D - dynamic
 0    ;;; Bloqueia VPC de acessar IPs dos Routers
      chain=forward action=drop src-address=192.168.10.0/24
      dst-address=!100.100.1.254 dst-address-list=IPS log=no log-prefix=""

 1    ;;; Bloqueia VPC de acessar o R8 local
      chain=input action=drop src-address=192.168.10.0/24
      dst-address-list=IPS log=no log-prefix=""
```

## 10.Tabela de Endereçamento

A tabela mostra dois ranges livres, isso aconteceu devido há algumas alterações nos enlaces da infraestrutura do laboratório. Os endereços de Loopback seguem uma sequência lógica, indo de 1 a 13, atribuindo aos Routes respectivamente.

| **Sub-rede** | **Host 1** | **Host 2** | **Broadcast** |
|---|---|---|---|
| 100.100.1.0/30 | 100.100.1.1 (R1) | 100.100.1.2 (R6) | 100.100.1.3 |
| 100.100.1.4/30 | 100.100.1.5 (R11) | 100.100.1.6 (R9) | 100.100.1.7 |
| 100.100.1.8/30 | 100.100.1.9 (R5) | 100.100.1.10 (R3) | 100.100.1.11 |
| 100.100.1.12/30 | 100.100.1.13 (R3) | 100.100.1.14 (R2) | 100.100.1.15 |
| 100.100.1.16/30 | 100.100.1.17 (R6) | 100.100.1.18 (R2) | 100.100.1.19 |
| 100.100.1.20/30 | 100.100.1.21 (R2) | 100.100.1.22 (R10) | 100.100.1.23 |
| 100.100.1.24/30 | 100.100.1.25 (R2) | 100.100.1.26 (R13) | 100.100.1.27 |
| 100.100.1.28/30 | 100.100.1.29 (R1) | 100.100.1.30 (R2) | 100.100.1.31 |
| 100.100.1.32/30 | 100.100.1.33 (R1) | 100.100.1.34 (R5) | 100.100.1.35 |
| 100.100.1.36/30 | 100.100.1.37 *(Livre)* | 100.100.1.38 *(Livre)* | 100.100.1.39 |
| 100.100.1.40/30 | 100.100.1.41 (R4) | 100.100.1.42 (R6) | 100.100.1.43 |
| 100.100.1.44/30 | 100.100.1.45 (R4) | 100.100.1.46 (R7) | 100.100.1.47 |
| 100.100.1.48/30 | 100.100.1.49 (R7) | 100.100.1.50 (R8) | 100.100.1.51 |
| 100.100.1.52/30 | 100.100.1.53 (R8) | 100.100.1.54 (R1) | 100.100.1.55 |

| **Sub-rede** | **Host 1** | **Host 2** | **Broadcast** |
|---|---|---|---|
| 100.100.1.56/30 | 100.100.1.57 *(Livre)* | 100.100.1.58 *(Livre)* | 100.100.1.59 |
| 100.100.1.60/30 | 100.100.1.61 (R12) | 100.100.1.62 (R1) | 100.100.1.63 |
| 100.100.1.64/30 | 100.100.1.65 (R11) | 100.100.1.66 (R1) | 100.100.1.67 |
| 100.100.1.252/30 | 100.100.1.253 (R1) | 100.100.1.254 (Nginx) | 100.100.1.255 |

| **Range VPLS DHCP R1 R8 - - > 192.168.10.0/24** | | | |
|---|---|---|---|
| 192.168.10.0/24 | 192.168.10.1 (R8) | 192.168.10.252 (VPC-R8) | 192.168.10.255 |
| 192.168.10.0/24 | 192.168.10.1 (R8) | 192.168.10.253 (VPC-R1) | 192.168.10.255 |
| 192.168.10.0/24 | 192.168.10.1 (R8) | 192.168.10.254 (R1) | 192.168.10.255 |